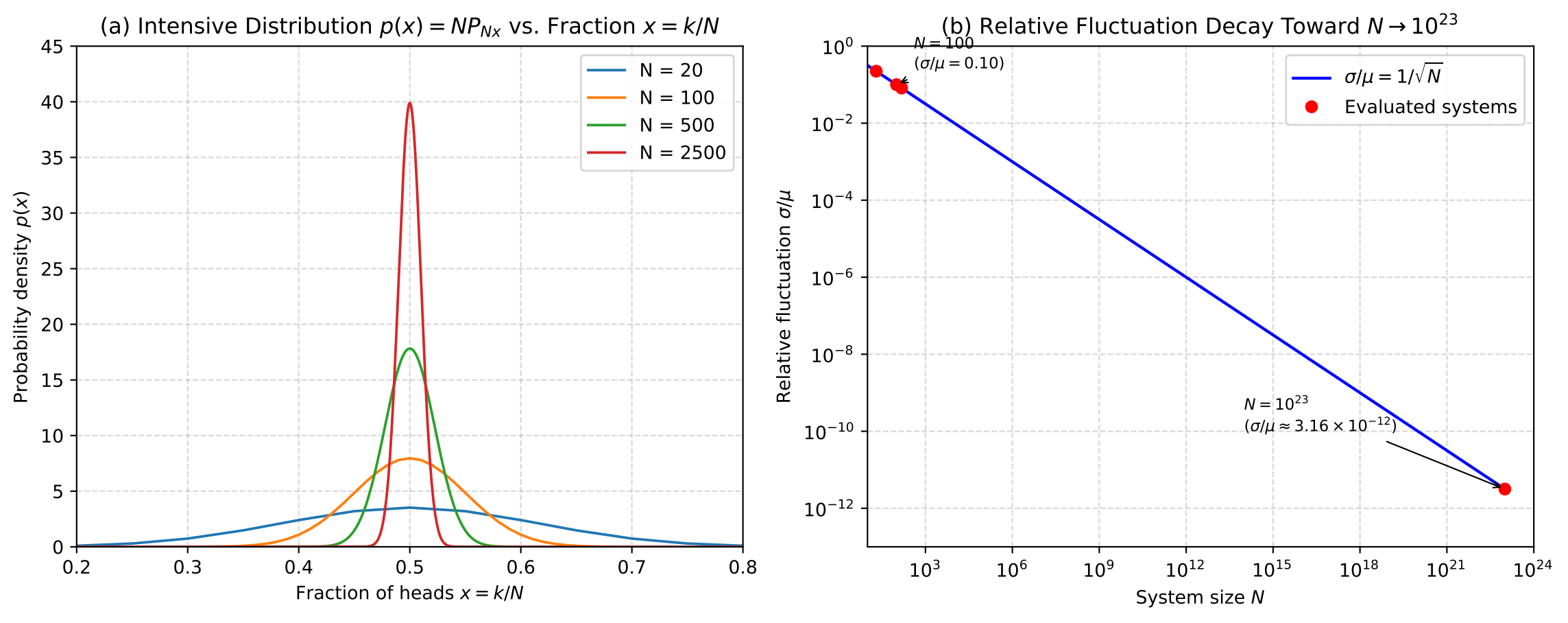
(a) Intensive Distribution $p(x) = NP_{Nx}$ vs. Fraction $x = k/N$
Probability density $p(x)$
Fraction of heads $x = k/N$
N = 20
N = 100
N = 500
N = 2500
0
5
10
15
20
25
30
35
40
45
0.2
0.3
0.4
0.5
0.6
0.7
0.8
(b) Relative Fluctuation Decay Toward $N \to 10^{23}$
Relative fluctuation σ/μ
System size N
$\sigma/\mu = 1/\sqrt{N}$
Evaluated systems
$N = 100$
$(\sigma/\mu = 0.10)$
$N = 10^{23}$
$(\sigma/\mu \approx 3.16 \times 10^{-12})$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{3}
10^{6}
10^{9}
10^{12}
10^{15}
10^{18}
10^{21}
10^{24}